The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
       k=k+1
       sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided ${0.4, 0.6}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 92 seconds

Yes, the point (0.4, 0.6) is at most 0.45 away from a minimizer in the infinity norm.

Explanation:

- The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$ as $n \to \infty$.
- $f(n)$ sums $\left(\frac{1}{2}\right)^k$ from $k = 3$ onward (total limit $\frac{1}{4}$) plus astronomically small positive

terms. Thus $a_2 = 1 + \delta$ with $\delta > 0$ unimaginably tiny.

- The problem is: minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1,\ x_1, x_2 \geq 0$. With $a_2 > 1$, the unique minimizer is $(x_1, x_2) = (0,\ 1/a_2) \approx (0, 1)$.
- The infinity-norm distance from $(0.4, 0.6)$ to $(0, 1/a_2)$ is $\max(0.4,\ |0.6 - 1/a_2|) = 0.4$ (since $1/a_2$ is exceedingly close to 1, making the second coordinate difference < 0.4).
- $0.4 \leq 0.45$, so the provided point is within the specified tolerance.